MIL-H-62315B(AT)
3 December 1990
SUPERSEDING
MIL-H-62315A(AT)
7 June 1985

MILITARY SPECIFICATION

HEATER ASSEMBLY, COMBUSTION VEHICULAR COMPARTMENT, 60,000 BTU/HR

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, design, testing, manufacturing, and acceptance requirements for a forced air inlet personnel heater assembly, referred to herein as "heater" (see 6.1).

1.2 Classification. Heaters shall be dual or single air of the following types:

Type I - Extended operating range.

Type II - Standard operating range.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2540

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:30Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS
FEDERAL

QQ-P-416	- Plating, Cadmium (Electrodeposited).
GG-P-455	- Plates and Foils, Photographic (Photosensitive Anodized Aluminum).
TT-C-490	- Cleaning, Methods, for Ferrous Surfaces and Pretreatment for Organic Coatings
VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-W-8611	- Welding, Metal Arc and Gas, Steels, and Corrosion and Heat Resistant Alloys; Process For
MIL-P-14105	- Paint, Heat Resisting (for Steel Surfaces).
MIL-P-19834	- Plates, Identification or Instruction, Metal Foil, Adhesive Backed.
MIL-S-22473	- Sealing, Locking, and Retaining Compounds: (Single Component).
MIL-C-22750	- Coating, Epoxy-Polyamide.
MIL-S-46163	- Sealing, Lubricating, and Wicking Compound: Thread-Locking, Anaerobic, Single-Component
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-P-53022	- Primer, Epoxy Coating, Corrosion Inhibiting, Lead and Chromate Free
MIL-P-53030	- Primer coating, Epoxy, Water Reducible, Lead and Chromate Free.
MIL-C-53039	- Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.
MIL-C-81562	- Coatings, Cadmium, Tin Cadmium

Source: https://assist.dla.mil -- Downloaded: 202610-23T01:30Z
Check the source to verify that this is the current version before use.

and Zinc (Mechanically Deposited)

STANDARDS
FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-45662	- Calibration System Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issue are those cited in the solicitation.

DRAWINGS
ARMY

11669489	- Heater, Vehicular Compartment
11669490	- Heater, Vehicular Compartment

(Copies of drawings required by manufacturers in connection with specific acquisition functions should be obtained from the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:30Z
Check the source to verify that this is the current version before use.

contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Performance Test Code - Application Part II of Fluid Meters (PTC) 19.5 Sixth Edition 1971 (Interim Supplement 19.5 on Instruments and Apparatus).

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 E. 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B633 - Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel

ASTM D2156 - Smoke Density in Flue Gases from Burning Distillate Fuels, Test for.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AIR MOVEMENT AND CONTROL ASSOCIATION (AMCA)

210-74 - Laboratory Methods of Testing Fans for Rating.

(Application for copies should be addressed to the Air Movement and Control Association, 30 W. University Drive, Arlington Heights, IL 60004.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

Source: https://assist.dla.mil -- Downloaded: 202610-23T01:30Z
Check the source to verify that this is the current version before use.

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish heaters which shall be subjected to first article inspection (see 4.4 and 6.3). First article inspection samples, properly marked with identifying information shall be representative of the units to be furnished to the Government. All subsequent heaters delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein, in applicable specifications, or drawings and shall be asbestos free. Materials not specifically designated shall be suitable for use in heaters operating as specified herein without any change in properties that would result in operation of the units falling outside of the specified limits (see 6.5). Recycled materials is acceptable when processed to make new.

3.2.1 Dissimilar metals. Except where necessary to complete an electrical circuit, the compatibility of dissimilar metals shall be in accordance with MIL-STD-889 (see 4.6.1).

3.2.2 Steel components. Non-painted steel components, other than stainless steel, shall be cadmium plated in accordance with QQ-P-416, type II, class 2 or MIL-C-81562, type II, class 2, or zinc plated according to ASTM B633, type II, class Fe/Zn 13 or MIL-C-81562, type II, class 5. Heat radiating surfaces may be aluminized steel (see 4.6.1).

3.2.3 Aluminum components. Aluminum components shall be coated in accordance with MIL-C-5541, class 1A (see 4.6.1).

3.2.4 Identification plates. Nameplates shall be furnished in one of the configurations as specified below (see 4.6.1):

a. Material and construction shall conform to MIL-P-514, type III, composition A, class 2, or composition C. Composition C material shall conform to GG-P-455, type II, grade A, class 1. Nameplate shall be permanently attached with metal fasteners.

b. Adhesive backed aluminum foil shall conform to MIL-P-19834, type I, glossy: except backing sheet, which may be made of paper, shall be removable intact by stripping without use of water or other solvents. Plates shall have and average adhesion, over bare and painted metal surfaces, of not less than 50 ounces per inch of width at 77 $\pm$ 5 degrees Fahrenheit (F).

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:30Z
Check the source to verify that this is the current version before use.

3.2.5 Welding. All heater parts that are exposed to combustion gases during normal operation shall be welded, as applicable, according to MIL-W-8611.

3.2.6 Sealant. All sealing threaded fittings in the heater's fuel circuit or threads utilizing a locking compound shall be sealed or locked, as applicable, with compound conforming to MIL-S-22473 or MIL-S-46163. Compound's grade shall be selected so that its strength shall not result in part damage upon disassembly for repair.

3.3 Design and construction. Heaters shall be constructed to the form and dimensions depicted on the applicable drawing specified by the contracting authority (see table I, 4.5.2, 4.6.2 and 6.2). The heater shall be fabricated from suitable corrosion resistant materials or treated to prevent corrosion. The heater combustion chamber, or heat exchanger, shall be made of stainless steel which is consistent with the life, corrosion resistance, and other design considerations of the assembly (see 4.6.1).

TABLE I. APPLICABLE DRAWINGS

Dual Air Heater

Type	Ordnance Drawing Number
I	11669489
II	11669490

3.3.1 Weight. Weight of the heater shall not exceed 38 pounds (lb) (see 4.6.3).

3.3.2 Electrical components and wiring. The heater's electrical components and wiring shall operate and meet all environmental requirements. The wire and insulation shall be flexible to -70 degrees F and be sized in accordance with the required rating (see 4.6.1).

3.3.3 Interchangeability. The assembly shall be so designed and fabricated that all component assemblies and parts are functionally and dimensionally interchangeable with like component assemblies and parts previously furnished by the same manufacturer. Construction shall be such that any part, except those furnished in matched sets or for which a selective fit is specified, may be installed, replaced, and adjusted without requiring modification (see 4.6.1).

3.3.4 Safety. Hot spots that could create contact burn to personnel shall be well insulated or guarded, or fall within the area identified as "hot spot" by the applicable drawing(s). The

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:30Z
Check the source to verify that this is the current version before use.

heater shall be so designed that, with proper installation, the heater exhaust gases cannot enter the vehicle personnel compartment (see 4.6.1).

3.4 Performance. The heater shall meet the performance requirements specified herein when mounted in any position from horizontal to vertical with the air inlet level with, or higher than, the air outlet; and, except as otherwise specified, when operating in both the high and low heat modes. The heater shall perform as specified using fuel conforming to VV-F-800 as follows:

Above plus 20 degrees F	type DF-2
Plus 20 to minus 25 degrees F	type DF-1
Below minus 25 degrees F	type DF-A

3.4.1 Fuel.

3.4.1.1 Fuel consumption. With any fuel pressure between 3 and 15 pounds per square inch gage (psig) at the fuel valve inlet, fuel consumption shall not exceed the following (see 4.6.4.1):

High heat	0.092 pounds per minute (lb/min)
Low heat	0.052 lb/min

3.4.1.2 Fuel leakage. There shall be no visible leakage from the heater fuel components with the heater operating and 15 psig fuel pressure applied to the inlet (see 4.6.4.2).

3.4.2 Voltage. The heater shall start and except where otherwise specified, meet performance requirements with an applied voltage of 19 to 30 volts (V) direct current (dc) (see 4.6.5).

3.4.3 Current. At 24 V dc, the current draw from the power source shall not exceed 16 amperes (A) starting and 15 A running when the fuel preheaters are not activated. Current draws may increase by 5 A when the fuel preheaters or start circuit are activated (at approximately 45 degrees F) (see 4.6.6).

3.4.4 Heat output. With a ventilating back pressure 1.30 inch (in) of water and an exhaust pressure of 0.95 (type I) or 0.85 (type II) in of water set as specified (see 3.4.5), the heater output shall be as follows (see 4.6.7):

High heat	19.0 to 30.0 V dc, 60,000 British Thermal Units per hour (BTU/hr) ± 10 percent. (19.0 ± 0.5 V dc optional at temperatures between 55 and 91

Source: https://assist.dla.mil -- Downloaded: 202610-23T01:30Z
Check the source to verify that this is the current version before use.

degrees F.)

Low heat	19.0 to 30.0 V dc, 30,000 BTU/hr ± 20 percent

3.4.5 Air discharge rate. With 24 V dc applied, the ventilating air discharge rate, based on a standard air density of 0.075 pounds per cubic foot (lb/cu ft) at 29.92 in of Mercury (Hg), 70 degrees F, into the specified restriction shall be as follows (see 4.6.8):

Pressure Minimum (in water)		Discharge rate Minimum (standard cubic feet per minute, scfm)
Ventilation	Exhaust	
	Type I	
1.30	0.95	165
	Type II	
0.00	0.00	200
1.30	0.85	150

3.4.6 Overheat switch. When subjected to the specified test, the heater shall meet the requirements for the applicable type (see 4.6.9):

Type I. The heater shall be equipped with an overheat switch to reduce the fuel flow to control the temperature of the discharge air so that it falls between 275 and 350 degrees F. When the air restriction is removed the heater shall continue burning without requiring a return to the start cycle. Upon further restriction of air flow to not greater than 75 scfm the ventilating air temperature shall not raise above 475 degrees F and the heater shall shut off its fuel flow then enter its purge cycle and shut itself off.

Type II. The heater shall be equipped with an overheat switch and an automatic fuel shut off device which will automatically open a circuit and shut off the fuel flow when the temperature of the discharge air reaches 400 ± 75 degrees F. Subsequently, it shall enter its purge cycle and then shut itself off.

3.4.7 Smoke density. After the heater has been operating on high heat for 5 minutes, the exhaust gases shall not exceed a smoke spot number of three below 55 degrees F and five above 55 degrees F as defined in ASTM D2156 (see 4.6.10).

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:30Z
Check the source to verify that this is the current version before use.

3.4.8 Starting time. The starting time, measured from the time the heater is turned on until the igniter circuit is de-energized, shall not exceed the following for the specified types (see 4.6.11):

Type I. Two minutes for temperatures above minus 20 degrees F. Four minutes for temperatures below minus 20 degrees F.

Type II. Four minutes.

3.4.9 Purging time. The purging time, measured from the time the heater switch is turned off until the blowers shut off at which time the exhaust temperature shall not be greater than 350 degrees F, shall not exceed 4.0 minutes (see 4.6.12).

3.4.10 Toxic fumes. The heater shall be designed so that the combustion products do not contaminate the ventilating air during operation of the heater (see 4.6.13).

3.4.11 Endurance. The heater shall meet the requirements of 3.4.1.1 and 3.4.2 through 3.4.7 and 3.4.10 after 800 hours of operation, and 3.4.1.2 and 3.4.8 and 3.4.9 during 800 hours of operation; including at least 2400 start cycles. The following maintenance is permissible (see 4.6.14):

a. Ice may be removed from the heater fuel system.
b. Cleaning and adjustment of the flame detector switch, but not more than two times.
c. Replacement of the igniter, but not more than one time.
d. Replacement of the burner assembly if required, but no more frequently than once per sample during the test procedure.

3.4.12 Environmental conditions. The heater shall meet the requirements of 3.4.1 through 3.4.10 after being subjected to the environmental conditions specified in 3.4.12.1 through 3.4.12.6.

3.4.12.1 High temperature. The heater shall meet the requirements after storage at 160 degrees F and during operation at temperatures up to 75 degrees F (see 4.6.15.1).

3.4.12.2 Low temperature. The heater shall meet the requirements after storage at a temperature of -65 degrees F and during operation at -55 degrees F provided the fuel is in liquid form (see 4.6.15.2).

3.4.12.3 Humidity. The heater shall meet the requirements after exposure to relative humidity up to 100 percent (see 4.6.15.3).

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:30Z
Check the source to verify that this is the current version before use.

3.4.12.4 Salt fog. The heater shall meet the requirements after exposure to salt fog for 96 hours (see 4.6.15.4).

3.4.12.5 Shock. The heater shall meet the requirements after being subjected to 40 gravity units (g) shock loading in three principles axis. The input is to be measured at the interface of the heater and support bracket (see 4.6.15.5).

3.4.12.6 Vibration. The heater shall meet the requirements during and after sinusoidal vibration in accordance with figure 1 herein. The input is to be measured at the interface of the heater and support bracket (see 4.6.15.6).

3.4.12.7 Electromagnetic Interference (EMI). The heater shall meet the requirements of MIL-STD-461A, Notice 4 (CE-04 and RE-02, narrow band optional) for all operating modes including the start cycle. However, the single switching actions which initiates and ends, respectively, the heater start cycle, do not have to meet the EMI requirements (see 4.6.15.7).

3.5 Protective Coating. All steel parts, except stainless steel, that are subject to temperatures exceeding 300 degrees F, shall be coated, color Forest Green or Olive Drab, per MIL-P-14105, 1.5 to 3.5 mils dry film thickness. Surfaces shall be free of oil, grease, rust and mill scale before treatment. All other painted steel parts shall be coated as described above or as follows: Clean and treat per Type I or III, of TT-C-490; Prime per MIL-P-53030, or MIL-P-53022, dry film thickness 0.8 mils minimum. Topcoat color shall be camouflage green, chip number 34094 of FED-STD-595, conforming to MIL-C-53039, MIL-C-22750, or MIL-C-46168. Total dry film thickness of both coats shall be between 2.6 and 4.5 mils.

3.6 Identification and marking. The heaters shall be marked in accordance with MIL-STD-130 (see 4.5.2 and 4.6.2).

3.7 Workmanship. The heater shall be free of defects such as burrs, scratches, chips, sharp edges, corrosion, scale and dirt. Workmanship shall also be in accordance with requirement 9 of MIL-STD-454 (see 4.5.2 and 4.6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:30Z
Check the source to verify that this is the current version before use.

witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to the acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the supplier is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
    1. Examination (see 4.5.2).
    2. Acceptance tests (see 4.5.3.1).
c. Control tests (see 4.5.3.2).

4.3 Inspection conditions. Unless otherwise specified herein or by the procuring agency (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 73 ± 18 degrees F
b. Barometric pressure 28.0 ± 2.5 in Hg
c. Relatively humidity 50 ± 40 percent

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select the first and two additional heaters produced under the production contract (see 6.2) for first article inspection. First article samples shall be numbered and inspected as specified in 4.4.1 and in table II. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply heaters that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:30Z
Check the source to verify that this is the current version before use.

contracting officer. The listing in Table II does not imply any particular test sequence.

4.4.1 In-process inspections. During fabrication of the first production heater, an in-process inspection shall be conducted by the contractor under the surveillance of the Government inspector. Inspection shall be made at the contractor's facility prior to application of primer and paint. Processing and welding procedures, quality system and inspection records shall be reviewed and evaluated during this inspection.

4.4.2 Production sample. After completion of inspection, the first production heater shall remain at the manufacturing facility, as a production sample, and be the last heater shipped on the contract. The heater may be released sooner at the discretion of the contracting officer. The contractor shall service and maintain the heater during this period.

TABLE II. Classification of inspections.

Title	Requirement	Test	First article Samples			Quality conformance A		C
			1	2	3	100%		
Heater position			H	V	H		H	
Materials and construction	3.2 thru 3.2.4 and 3.3.2 thru 3.3.4	4.6.1	X	X	X			X
Defects (see Table III)	3.3, 3.5, 3.6 & 3.7	4.6.2	X	X	X			X
Weight	3.3.1	4.6.3	X	X	X			X
Fuel consumption	3.4.1.1	4.6.4.1	X	X	X	X	X	
Fuel leakage	3.4.1.2	4.6.4.2	X	X	X	X		X
Voltage	3.4.2	4.6.5	X	X	X	X		X
Current	3.4.3	4.6.6	X	X	X	X		X
Heat output	3.4.4	4.6.7	X	X	X			X
Air discharge rate	3.4.5	4.6.8	X	X	X	X		X
Overheat switch	3.4.6	4.6.9	X	X	X	X		X
Smoke density	3.4.7	4.6.10	X	X	X	X		X
Starting time	3.4.8	4.6.11	X	X	X	X		X
Purging time	3.4.9	4.6.12	X	X	X	X		X
Toxic fumes	3.4.10	4.6.13	X	X	X			X
Endurance	3.4.11	4.6.14	X	X				

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:30Z
Check the source to verify that this is the current version before use.

Endurance (reduced)	3.4.11	4.6.14.1					X
High temperature	3.4.12.1	4.6.15.1	X	X	X		X
Low temperature	3.4.12.2	4.6.15.2	X	X	X		X
Humidity	3.4.12.3	4.6.15.3		X			
Salt fog	3.4.12.4	4.6.15.4	X				
Shock	3.4.12.5	4.6.15.5	X	X			X
Vibration	3.4.12.6	4.6.15.6		X	X		X
EMI	3.4.12.7	4.6.15.7		X	X		

H=Horizontal V=Vertical A=Acceptance C=Control

4.4.3 First article inspection failure. The procuring agency shall determine whether a particular first article inspection failure shall be treated as major or minor. A major failure is a noncompliance with requirements specified herein by a functioning heater component, heat exchanger or requires corrective action that involves a design change. At least the endurance or environmental test which caused the deficiency shall be repeated. A complete first article test repeat of any or all units may be necessary at the discretion of the procuring agency. A minor failure may result in, at most, a repeat of the endurance or environmental test which caused the deficiency and any related test at the discretion of the procuring agency. The Government shall not accept delivery of any units until first article test requirements as specified in the contract have been satisfied.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the heaters of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105. Before sampling may be initiated, the contractor shall 100 percent inspect the first 100 heaters to establish a process average (see 6.4) to allow normal sampling in accordance with MIL-STD-105.

4.5.2 Quality conformance examinations.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:30Z
Check the source to verify that this is the current version before use.

conformance to the following acceptable quality levels (AQL) on the basis of percent defective.

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table III.

TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Assembly, incomplete (see 3.3).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.3).	SIE 1/
103	Identification marking, improper (see 3.6).	Visual
104	Faulty workmanship affecting performance (see 3.7).	Visual
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3).	SIE 1/
202	Protective coating, improper (see 3.5).	Visual
203	Faulty workmanship affecting appearance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Quality conformance tests.

4.5.3.1 Acceptance test (100 percent). Each heater shall be subjected to the acceptance test specified in table II.

4.5.3.2 Control tests. Control tests shall be conducted on one heater from each lot of 600 units consecutively produced, except that not more than one test shall be performed in a three month period, nor less than one test in a six month period. The heaters shall be subjected to the control tests specified in table II.

4.5.3.3 Failure. Failure of any heater to pass any of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:30Z
Check the source to verify that this is the current version before use.

specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.6 Method of inspection. Except as otherwise specified herein, performance tests shall be conducted with the heater mounted in the positions specified in table II, at both high and low heat, and at input voltages of 19.0 ± 0.5 V dc, 24.0 ± 0.5 V dc and 30.0 ± 0.5 V dc. Acceptance testing shall be conducted at 24.0 ± 0.5 V dc only. The fuel shall be as specified in 3.4 and the fuel pressure shall be 8 ± 1 psig. Prior to each start, the heater shall be cooled for at least 10 minutes.

4.6.1 Materials and construction. To determine conformance to 3.2 through 3.2.4, and 3.3.2 through 3.3.4, inspection and material certification records shall be maintained by the contractor. Records shall be subject to review by the Government and shall include date, part, or characteristic identification, inspection results, and disposition of lot (accepted or rejected). Corrective action taken on noted defects shall be subject to approval by the Government.

4.6.2 Defects. Conformance to 3.3, 3.5, and 3.7, shall be determined by examination for the defects listed in table III. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.6.3 Weight. To determine conformance to 3.3.1, the heaters shall be weighed.

4.6.4 Fuel.

4.6.4.1 Fuel consumption. To determine conformance to 3.4.1.1 the fuel flow shall be measured with a fuel pressure of 15 (+0, -1) psig applied to the fuel valve inlet and subsequently with a pressure of 3 (+1, -0) psig, similarly applied.

4.6.4.2 Fuel leakage. To determine conformance to 3.4.1.2 the heater shall be operated for at least 5 minutes at a fuel pressure of 14.0 ± 1 psig. Subsequently, there shall be no evidence of fuel leakage.

4.6.5 Voltage. To determine conformance to 3.4.2, applied voltage shall be monitored throughout performance testing.

4.6.6 Current. To determine conformance to 3.4.3, current draw from the power source shall be monitored during performance testing. Running current shall be measured after the igniter or start circuit is de-energized.

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:30Z
Check the source to verify that this is the current version before use.

4.6.7 Heat output. With ventilating and exhaust back pressures set as specified (see 4.6.8), thermocouples measuring input and output air temperatures, and measuring ventilating air flow (see 4.6.8), the heater shall be operated until the ventilating outlet temperature has stabilized. Output thermocouples shall be located 42 ± 1 in from the heater, within the duct used in 4.6.8. When testing is performed at ambient temperatures above 55 degrees F, and the overheat switch cycles, the switch may be bypassed. Bypassing is only allowed when the outlet temperature reaches between 275 and 350 (type I heaters) or 400 ± 75 (type II heaters) degrees F. Heat output shall be calculated, using the following equation, to determine conformance to 3.4.4.

H = V x d x Cp x T
H = Heat output in BTU/hr.
V = Air flow in cubic feet per hour (cu ft/hr) (standard air).
d = Density of air at 70 degrees F, 0.0750 lb/cu ft. (standard air).
Cp = Specific heat of air, 0.24 BTU per lb per degree F, as corrected for moisture content and temperature.
T = Outlet temperature, in degrees F, minus inlet temperature, in degrees F.

4.6.8 Air discharge rate. To determine conformance to 3.4.5, operate the heater as necessary and set the ventilating air restriction to produce a minimum of 1.30 in water pressure drop at 174 ± 5 scfm air flow. Measure the air flow and pressure drops according to AMCA 210-74 or ASME PTC 19.5. With the heater operating at 24 V dc input, set the exhaust restriction as specified and measure the ventilating air flow rate and back pressures of the exhaust and ventilating air as specified. During subsequent testing no changes shall be made in these restrictions. The ventilating air duct shall be insulated with at least 1.5 in. of fiberglass.

4.6.9 Overheat switch. To determine conformance to 3.4.6, expose the heater to a temperature between 55 and 91 degrees F and operate it at high heat into the duct used in 4.6.8, restricting the ventilating air flow as necessary to cause a temperature buildup in the duct. Measure the temperature of the ventilating air at a point 42 ± 1 in from the heater, within the duct. The overheat switch shall break the circuit and reduce or cut off the fuel when the air in the duct reaches the temperature specified for its type. The types shall function as follows:

Type I - Over heat temperature is between 275 and 350 degrees F. After actuation of the over heat function, the heater shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:30Z
Check the source to verify that this is the current version before use.

operated for at least two minutes and measure the control temperature. Upon removing the restriction the heater shall resume burning without requiring return to the start cycle. Subsequently, the ventilating air shall be further restricted and the heater's performance shall be monitored for conformance.

Type II - Over heat temperature is 400 ± 75 degrees F. The heater shall be monitored for required performance.

NOTE: Heater radiation effects should be negated by appropriate shielding of heat sensing devices.

4.6.10 Smoke density. To determine conformance to 3.4.7, using the test methods set forth in ASTM D2156, measure the smoke density of the heater exhaust gases after 5 minutes on high heat. The smoke density shall not exceed a smoke number of three below 55 degrees F and five above 55 degrees F at 24 V dc.

4.6.11 Starting time. To determine conformance to 3.4.8, measure and record the time from the starting switch being turned on until the igniter, start circuit, or preheaters are de-energized. The elapsed time shall not exceed the value specified herein. This test shall be conducted at low heat only.

4.6.12 Purging time. To determine conformance to 3.4.9, the heater's ventilating air and exhaust temperatures shall be stabilized at high heat and the starting switch shall be turned off. Measure the exhaust temperature (not more than 25 in from the heater's exhaust tube's end) and the time from the starting switch being turned off until the blowers shut off. The elapsed time shall not exceed 4.0 minutes and exhaust temperature not more than 350 degrees F.

4.6.13 Toxic Fumes.

4.6.13.1 Common intake heaters. Prior to heater assembly, all heat exchangers shall be tested for leakage with 10 psig applied under water. No leakage shall be permitted. The common intake heater need only be tested again at the completion of the first article and control tests, at which time leakage shall not exceed 0.5 scfm with 3 inches water pressure applied.

4.6.13.2 Dual intake heaters. Prior to heater assembly, all heat exchangers shall be tested for leakage with 10 psig applied under water. No leakage shall be permitted. Once heater is assembled, leakage shall not exceed one scfm with the combustion air inlet sealed and 3 inches water pressure applied to the

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:30Z
Check the source to verify that this is the currentversion before use.

exhaust outlet.

4.6.14 Endurance. To determine conformance to 3.4.11, the heater shall be operated with the specified restrictions (see 4.6.8) for a total of 800 hours "ON" time (ignite and run) (see 4.6.14.1). Operation shall be in 30 minute cycles consisting of 20 minutes "ON" time and 10 minutes "OFF" time (purge and cool). The total 800 hours of operation shall consist of only "ON" time. "OFF" time will not be included in compilation of operating time. During endurance testing the heater shall be tested as specified by 4.6.11 and 4.6.12. After endurance testing the heater shall be tested as specified by 4.6.13. Operation shall consist of the following phases:

PHASE I - Using DF-2 fuel at heater ambient temperature of 73 ± 18 degrees F,

a. Set controls at low heat and operate heater in cycles previously described for 8 hours with an input voltage of 19.0 ± 0.5 V dc.
b. Repeat step a using an input voltage of 24.0 ± 0.5 V dc.
c. Repeat step a using an input voltage of 28.0 ± 0.5 V dc.
d. Reset temperature control to high heat and repeat steps a, b and c.
e. Repeat steps "a" through "d" until 600 thirty minute cycles are completed (200 "ON" hours).

PHASE II - Using DF-1 fuel at minus 25 ± 5 degrees F (both fuel and heater) repeat steps "a" through "e", except 300 thirty minute cycles (100 "ON" hours) shall be completed and in Step "c" substitute 30 ± 0.5 V dc in lieu of 28.0 ± 0.5 V dc.

PHASE III - Using DF-A fuel at minus 55 ± 5 degrees F (both fuel and heater) repeat steps "a" through "e", except 300 thirty minute cycles (100 "ON" hours) shall be completed and in Step "c" substitute 30 ± 0.5 V dc in lieu of 28.0 ± V dc.

PHASE IV - Repeat phases I through III for an additional 400 "ON" hours. After the finial cycle at each specified temperature and still at that temperature the heater shall be tested as specified in 4.6.4 through 4.6.10 and 4.6.12.

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:30Z
Check the source to verify that this is the current version before use.

4.6.14.1 Reduced endurance testing. For control test purposes, the "ON" time endurance testing (see 4.6.14) shall be reduced to 150 hours. The test shall consist of 50 hours each of phases I, II and III.

4.6.15 Environmental. After the completion of environmental testing as specified in 4.6.15.1 through 4.6.15.6, the heater shall be subjected to the tests specified in 4.6.4 through 4.6.13.

4.6.15.1 High temperature. To determine conformance to 3.4.12.1, the heater shall be tested in accordance with MIL-STD-810, method 501, procedure I. The heater, positioned as specified in table II, shall be stored at 160 degrees F for 24 hours.

4.6.15.2 Low temperature. To determine conformance to 3.4.12.2, the heater shall be tested in accordance with MIL-STD-810, method 502, procedures I and II as follows:

a. Procedure I: The heater, using DF-A fuel, shall be positioned as specified in table II. The chamber air temperature shall be maintained at -65 degrees F for at least 12 hours. The heater shall immediately be tested as specified in procedure II.

b. Procedure II: The chamber temperature shall be raised to -55 degrees F, the operation temperature for DF-A fuel, and the heater shall then be tested as specified in 4.6.4 through 4.6.12, omitting 4.6.4.2 and 4.6.9. Prior to testing, the heater shall be stabilized at the low operation temperature. During testing, stabilize combustion for 30 minutes on low heat and again on high heat. Subsequently, testing as specified in procedure II, shall be repeated at the low operation temperature of minus 25 degrees F using DF-1 fuel and at plus 20 degrees F using DF-2 fuel. At each of the three temperatures, the tests specified in 4.6.11 and 4.6.12 shall be repeated twice more, for a total of three times. A baffle may be used to reduce cold air blasts effects for all cold room tests. The baffle must allow cold air flow around the heater. The combustion air inlet in multiport air heaters must be outside the baffle.

4.6.15.3 Humidity. To determine conformance to 3.4.12.3, the heater shall be operated for a minimum of one hour at high heat and then tested as specified in MIL-STD-810, method 507, procedure III.

4.6.15.4 Salt Fog. To determine conformance to 3.4.12.4, the heater shall be operated for a minimum of one hour at high heat

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:30Z
Check the source to verify that this is the current version before use.

and then tested in accordance with MIL-STD-810, method 509, procedure I. The heater shall be exposed to salt fog for a period of 96 hours.

4.6.15.5 Shock. To determine conformance to 3.4.12.5, the heater, with no fuel applied, shall be subjected to the half-sine shock pulse wave form shown by figure 2 herein according to the method specified in MIL-STD-810, method 516, procedure I. The pulse shall have an amplitude of 40 g's and a duration of 18 milliseconds. During testing, the heater shall be subjected to input power and monitored for intermittent operation. Steps 2 and 5 may be omitted.

4.6.15.6 Vibration. To determine conformance to 3.4.12.6, the heater exposed to temperatures between 55 and 91 degrees F shall be rigidly mounted and subjected to vibration as specified in MIL-STD-810, method 514, procedure VIII, using the curve shown by figure 1 herein. During testing, heater shall be started, purged, operated at both heats, and monitored for intermittent operation. A resonant frequency search shall be conducted along each axis, with the heater operating at low heat, over a range of 5 to 500 Hertz. The test schedule shall include 13-1/3 minutes dwell time at each resonance to a maximum of four. After determining resonance at low heat, the specified resonance dwell and sinusoidal cycling tests shall be conducted. The total cycling time per axis shall be 80 minutes including the resonance dwell time.

4.6.15.7 Electromagnetic Interference. The heater shall be subject to the tests specified by MIL-STD-462, notice 3.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

6.1 Item definition. The heater shall be a diesel fuel burning unit with an electrical control system and an electric blower.

6.1.1 Intended use. The heater furnished under this specification is intended to supply warm air to the crew compartment of military vehicles. Type I heaters may be substituted for type II heaters. Type II heaters shall not be substituted for type I heaters.

6.2 Ordering data: Acquisition documents should specify the following:

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:30Z
Check the source to verify that this is the current version before use.

a. Title, number, and date of this specification.
b. If first article samples are not required (see 3.1).
c. Title, number and date of applicable drawing (see 3.3).
d. If responsibility for inspection shall be other than as specified (see 4.1.1).
e. If responsibility for inspection equipment shall be other than as specified (see 4.1.1).
f. If inspection conditions shall be other than as specified (see 4.3).
g. If first article inspection is not required (see 4.4 and 6.3).
h. Selection of applicable level and packaging requirements (see 5.1).

6.3 First article. First article samples shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for examination, tests, and acquisition of the first article (see 3.1).

6.4 Criticality. This is a critical item.

6.5 Process average. Sampling may be initiated if the process average value for the first 100 heaters inspected is less than the AQL specified in the classification of defects for major and minor defects.

$$\text{Process average} = \frac{\text{Number of defects}}{\text{Number of heaters inspected}} \times 100$$

If the computed process average exceeds the specified AQL, 100 percent inspection shall be performed and continued until such time that the process average for 100 consecutive heaters is less that the specified AQL (see 4.5.1.2).

6.6 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications, without jeopardizing the intended use of the item, shall be encouraged.

6.7 Changes from previous issue. Asterisks, to identify changes with respect to the previous issue, are not used in this revision due to the extensiveness of the changes.

Custodian:
Army - AT:

Preparing activity:
Army - AT

(Project No. 2540-A028)

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:30Z
Check the source to verify that this is the current version before use.

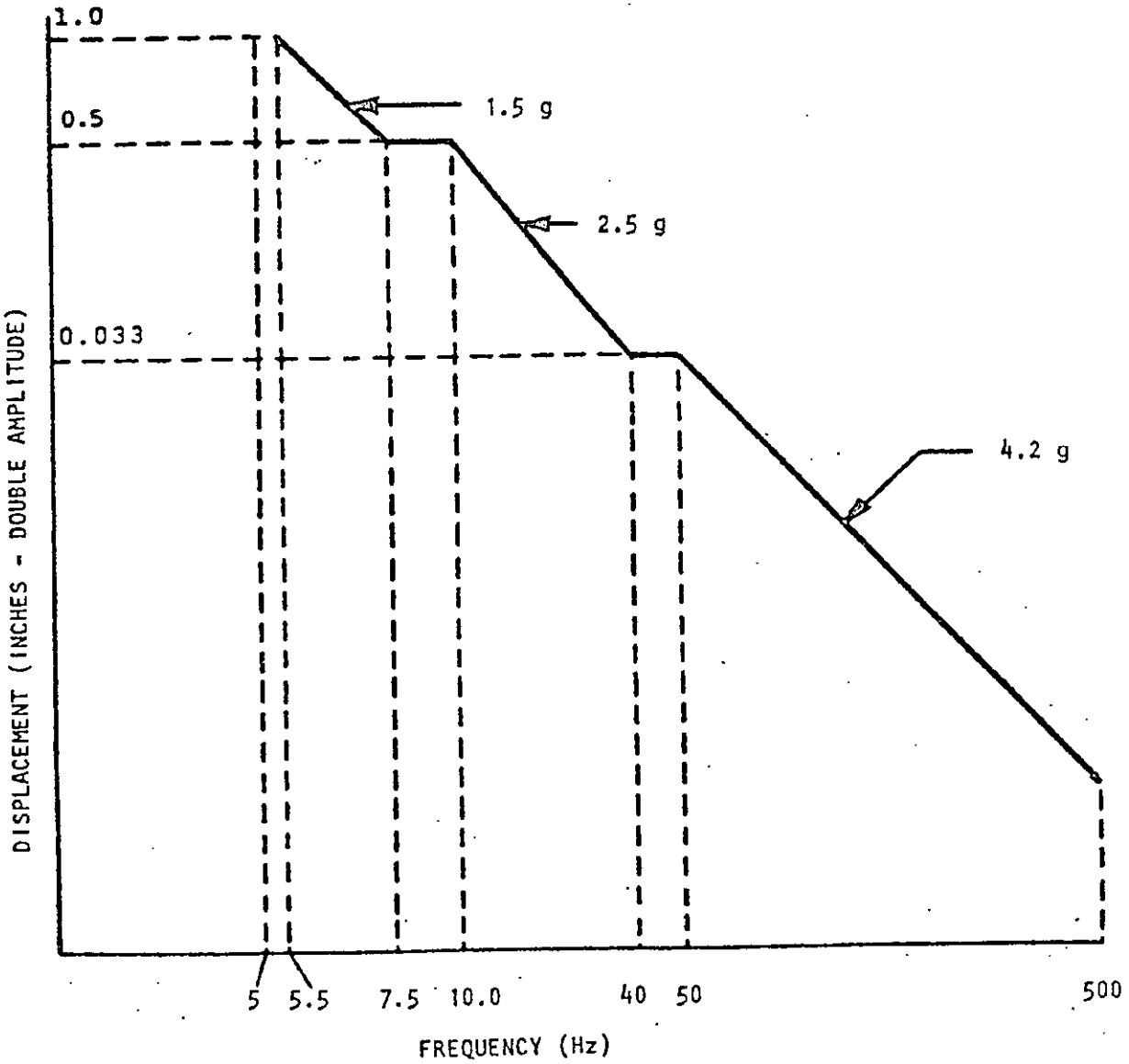
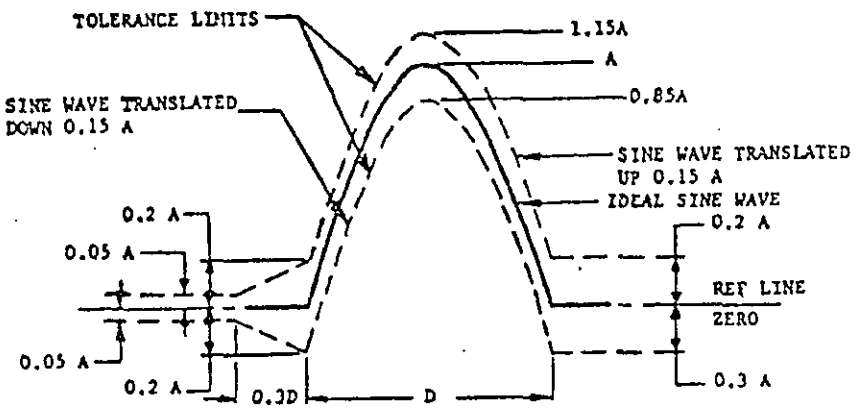


FIGURE 1. Vibration curve.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:30Z
Check the source to verify that this is the current version before use.



NOTE: THE OSCILLOGRAM SHALL INCLUDE A TIME ABOUT 3D LONG WITH A PULSE LOCATED APPROXIMATELY IN THE CENTER. THE ACCELERATION AMPLITUDE OF THE IDEAL HALF-SINE PULSE IS A AND ITS DURATION IS D. THE MEASURED ACCELERATION PULSE SHALL BE CONTAINED BETWEEN THE BROKEN LINE BOUNDARIES AND THE MEASURED VELOCITY CHANGE (WHICH MAY BE OBTAINED BY INTEGRATION OF THE ACCELERATION PULSE) SHALL BE WITHIN THE LIMITS $V_i \pm 0.1\ V_i$ WHERE V_i IS THE VELOCITY-CHANGE ASSOCIATED WITH THE IDEAL PULSE WHICH EQUALS $2AD/\pi$. THE INTEGRATION TO DETERMINE VELOCITY CHANGE SHALL EXTEND FROM 0.4D BEFORE THE PULSE TO 0.1D AFTER THE PULSE.

FIGURE 2. Half-sine shock pulse configuration and its limits.

Source: https://assist.dla.mil -- Downloaded: 202610-23T01:30Z
Check the source to verify that this is the current version before use.